

Doctors :-- doctor_id doctor_name specialization experience department

1	Dr. Smith	Cardiologist	10	Cardiology
2	Dr. John	Neurologist	8	Neurology
3	Dr. Alex	Dermatologist	6	Skin
4	Dr. Mary	Orthopedic	12	Bone
5	Dr. Rahul	Pediatrician	7	Child Care

Appointments :-- appointment_id patient_name doctor_id

appointment_date fees

101	Alice	1	2024-01-02	500
102	Bob	2	2024-01-03	600
103	Charlie	2	2024-01-05	600
104	David	6	2024-01-06	700
105	Emma	3	2024-01-07	400
106	Frank	NULL	2024-01-09	350

1. Write a query to display all doctors and their corresponding appointment details using RIGHT JOIN.

select doctor_name,appointment_id,patient_name,appointment_date,fees

from appointments

right join doctors

using(doctor_id);

	doctor_name	appointment_id	patient_name	appointment_date	fees
▶	dr. smith	101	alice	2024-01-02	500
	dr. john	103	charlie	2024-01-05	600
	dr. john	102	bob	2024-01-03	600
	dr. alex	105	emma	2024-01-07	400
	dr. mary	NULL	NULL	NULL	NULL
	dr. rahul	NULL	NULL	NULL	NULL

2. Show each doctor's name along with the patient's name. Include doctors who have no appointments.

```
select doctor_name,patient_name
from appointments
right join doctors
using(doctor_id);
```

	doctor_name	patient_name
▶	dr. smith	alice
	dr. john	charlie
	dr. john	bob
	dr. alex	emma
	dr. mary	NULL
	dr. rahul	NULL

Retrieve doctor name, specialization, and appointment date for all doctors, even those without patients.

```
select doctor_name,specialization,appointment_date
from appointments
right join doctors
using(doctor_id);
```

	doctor_name	specialization	appointment_date
▶	dr. smith	cardiologist	2024-01-02
	dr. john	neurologist	2024-01-05
	dr. john	neurologist	2024-01-03
	dr. alex	dermatologist	2024-01-07
	dr. mary	orthopedic	NULL
	dr. rahul	pediatrician	NULL

Find the list of doctors who don't have any appointments yet.

```
select doctor_name
from appointments
right join doctors
using(doctor_id)
where appointment_id is null;
```

	doctor_name
▶	dr. mary
	dr. rahul

Display each doctor's name and the total number of appointments they have (include 0 if none).

```
select doctor_name,count(appointment_id) as totalNoOfAppointments
from appointments
right join doctors
using(doctor_id)
group by doctor_name;
```

	doctor_name	totalNoOfAppointments
▶	dr. smith	1
	dr. john	2
	dr. alex	1
	dr. mary	0
	dr. rahul	0

Show the total fees collected by each doctor (display 0 if no fees were collected).

```
select doctor_name,sum(fees) as totalFees
from appointments
right join doctors
using(doctor_id)
group by doctor_name;
```

	doctor_name	totalFees
▶	dr. smith	500
	dr. john	1200
	dr. alex	400
	dr. mary	NULL
	dr. rahul	NULL

List all doctors and the patients they treated, ordered by doctor name.

```
select doctor_name,patient_name
from appointments
right join doctors
using(doctor_id)
order by doctor_name;
```

	doctor_name	patient_name
▶	dr. alex	emma
	dr. john	charlie
	dr. john	bob
	dr. mary	NULL
	dr. rahul	NULL
	dr. smith	alice

Display doctor name, specialization, and average fee charged per doctor.

```
select doctor_name,specialization,avg(fees) as avgFees
from appointments
right join doctors
using(doctor_id)
group by doctor_name,specialization;
```

	doctor_name	specialization	avgFees
▶	dr. smith	cardiologist	500.0000
	dr. john	neurologist	600.0000
	dr. alex	dermatologist	400.0000
	dr. mary	orthopedic	NULL
	dr. rahul	pediatrician	NULL

Find doctors whose average consultation fee is greater than ₹500.

```
select doctor_name,avg(fees) as avgFees
from appointments
right join doctors
using(doctor_id)
group by doctor_name
having avgFees>500;
```

	doctor_name	avgFees
▶	dr. john	600.0000

Show all doctor names with their department and total number of **distinct patients**.

```
select doctor_name,department,count(distinct patient_name) as totalNoOfDistinctPatients
from appointments
right join doctors
using(doctor_id)
group by doctor_name,department;
```

	doctor_name	department	totalNoOfDistinctPatients
▶	dr. alex	skin	1
	dr. john	neurology	2
	dr. mary	bone	0
	dr. rahul	child care	0
	dr. smith	cardiology	1

List doctors who have treated **at least two unique patients**.

```
select doctor_name,count( distinct patient_name) as uniquePatients
from appointments
right join doctors
using(doctor_id)
group by doctor_name;
```

	doctor_name	uniquePatients
▶	dr. alex	1
	dr. john	2
	dr. mary	0
	dr. rahul	0
	dr. smith	1

Display each department and total revenue generated from its doctors (include departments with no revenue).

```
select department,sum(fees) as totalRevenue
from appointments
right join doctors
using(doctor_id)
group by department;
```

	department	totalRevenue
▶	cardiology	500
	neurology	1200
	skin	400
	bone	NULL
	child care	NULL

Retrieve each doctor's name and their **most recent appointment date** (if any).

```
select doctor_name,appointment_date
from appointments
right join doctors
using(doctor_id)
group by doctor_name,appointment_date
order by appointment_date desc;
```

	doctor_name	appointment_date
▶	dr. alex	2024-01-07
	dr. john	2024-01-05
	dr. john	2024-01-03
	dr. smith	2024-01-02
	dr. mary	NULL
	dr. rahul	NULL

Show all doctors, replacing missing patient names with "No Patient".

```
select doctor_name,ifnull(patient_name,"NO Patients") as P_name
from appointments
right join doctors
using(doctor_id);
```

	doctor_name	P_name
▶	dr. smith	alice
	dr. john	charlie
	dr. john	bob
	dr. alex	emma
	dr. mary	NO Patients
	dr. rahul	NO Patients

Find the **total number of appointments**, ensuring that doctors without appointments are also included.

```
select doctor_name,count(appointment_id) as totalAppointments
from appointments
right join doctors
using(doctor_id)
group by doctor_name;
```

	doctor_name	totalAppointments
▶	dr. smith	1
	dr. john	2
	dr. alex	1
	dr. mary	0
	dr. rahul	0